

ECO 310 Problem Set 2 - Problem 3 Solution

vNM Axiom Violation

Problem Statement

Dave has preferences over lotteries which assign probabilities (p_a, p_b, p_c) to three possible prizes: an apple, a banana, and a cherry.

Given information:

- Dave is **indifferent** between:
 - lottery $p = (1, 0, 0)$ (100% apple)
 - lottery $q = (0, \frac{1}{2}, \frac{1}{2})$ (50% banana, 50% cherry)
- Dave **strictly prefers**:
 - lottery $r = (\frac{1}{2}, \frac{1}{2}, 0)$ (50% apple, 50% banana)
 - over lottery $s = (0, \frac{3}{4}, \frac{1}{4})$ (75% banana, 25% cherry)

Goal: Show that Dave must violate at least one vNM axiom.

Solution

Step 1: Set up vNM utilities

If Dave's preferences satisfy the von Neumann-Morgenstern axioms, then they can be represented by an expected utility function. Let the prizes be $x_1 = \text{apple}$, $x_2 = \text{banana}$, $x_3 = \text{cherry}$, with utilities:

$$\begin{aligned}u(x_1) &= u(\text{apple}) \\u(x_2) &= u(\text{banana}) \\u(x_3) &= u(\text{cherry})\end{aligned}$$

Step 2: Use the indifference condition

Since Dave is indifferent between p and q , their expected utilities must be equal:

$$\begin{aligned}
EU(p) &= EU(q) \\
1 \cdot u(x_1) + 0 \cdot u(x_2) + 0 \cdot u(x_3) &= 0 \cdot u(x_1) + \frac{1}{2} \cdot u(x_2) + \frac{1}{2} \cdot u(x_3) \\
u(x_1) &= \frac{1}{2}u(x_2) + \frac{1}{2}u(x_3)
\end{aligned}$$

Multiplying both sides by 2:

$$\boxed{2u(x_1) = u(x_2) + u(x_3)} \quad (\text{Equation 1}) \tag{1}$$

Step 3: Use the strict preference

Since Dave strictly prefers r to s , we must have:

$$\begin{aligned}
EU(r) &> EU(s) \\
\frac{1}{2}u(x_1) + \frac{1}{2}u(x_2) + 0 \cdot u(x_3) &> 0 \cdot u(x_1) + \frac{3}{4}u(x_2) + \frac{1}{4}u(x_3) \\
\frac{1}{2}u(x_1) + \frac{1}{2}u(x_2) &> \frac{3}{4}u(x_2) + \frac{1}{4}u(x_3) \\
\frac{1}{2}u(x_1) &> \frac{3}{4}u(x_2) - \frac{1}{2}u(x_2) + \frac{1}{4}u(x_3) \\
\frac{1}{2}u(x_1) &> \frac{1}{4}u(x_2) + \frac{1}{4}u(x_3)
\end{aligned}$$

Multiplying both sides by 4:

$$\boxed{2u(x_1) > u(x_2) + u(x_3)} \quad (\text{Inequality 2}) \tag{2}$$

Step 4: Show the contradiction

From Equation 1: $2u(x_1) = u(x_2) + u(x_3)$

From Inequality 2: $2u(x_1) > u(x_2) + u(x_3)$

Contradiction! We cannot have $2u(x_1)$ both *equal to* and *strictly greater than* $u(x_2) + u(x_3)$ simultaneously.

Conclusion

Therefore, no vNM utility representation exists for Dave's preferences. This means Dave must violate at least one of the von Neumann-Morgenstern axioms.

Which Axiom is Violated?

The **Independence Axiom (A.3)** is violated. Here's why:

Notice that we can write r and s as mixtures with the same mixing probability. Let $t = (0, 1, 0)$ (the lottery giving banana for sure). Then:

$$\begin{aligned} r &= \frac{1}{2}p + \frac{1}{2}t = \frac{1}{2}(1, 0, 0) + \frac{1}{2}(0, 1, 0) = \left(\frac{1}{2}, \frac{1}{2}, 0\right) \\ s &= \frac{1}{2}q + \frac{1}{2}t = \frac{1}{2}(0, \frac{1}{2}, \frac{1}{2}) + \frac{1}{2}(0, 1, 0) = \left(0, \frac{3}{4}, \frac{1}{4}\right) \end{aligned}$$

The Independence Axiom states that for any lotteries p, q, t and any $\alpha \in (0, 1)$:

$$p \sim q \implies \alpha p + (1 - \alpha)t \sim \alpha q + (1 - \alpha)t$$

We're told that $p \sim q$. By Independence with $\alpha = \frac{1}{2}$, this should imply:

$$\frac{1}{2}p + \frac{1}{2}t \sim \frac{1}{2}q + \frac{1}{2}t$$

That is, $r \sim s$. But we're told that Dave strictly prefers $r \succ s$. This contradicts Independence.

Intuition

Dave's first preference states that an apple (prize x_1) is *exactly as good* as a 50-50 mix of banana and cherry. However, his second preference reveals that an apple must be worth *strictly more* than that same 50-50 mix. These two statements are logically incompatible under expected utility theory.